

Signed-cover descent for the witness (13; 4, 18, 468) at $q = 61$

$$h(r) = 4r^4 + r^3 + 35r^2 + 8r + 28, \quad (t_j) = (4, 13, 18, 41)$$

4

$h'(4) = 18$: nonsquare

$$6^2 = 2 \cdot 18$$

13

$h'(13) = 38$: nonsquare

$$25^2 = 2 \cdot 38$$

18

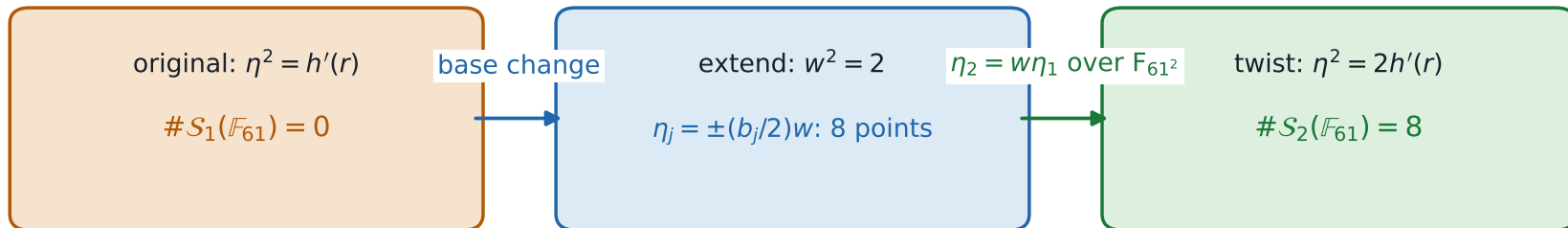
$h'(18) = 26$: nonsquare

$$28^2 = 2 \cdot 26$$

41

$h'(41) = 30$: nonsquare

$$11^2 = 2 \cdot 30$$



$w^{61} = -w$: Frobenius is four disjoint transpositions, $N_n = 0$ for odd n and 8 for even n .

$$Z_{S_1}(T) = (1 - T^2)^{-4}; \text{Tr}(\text{Frob}) = 0 \text{ but } \text{Tr}_{S/\mathbb{F}_{61}}(1) = 8.$$

The extension and the twist are exact typed constructions; neither deletes the original ES witness.